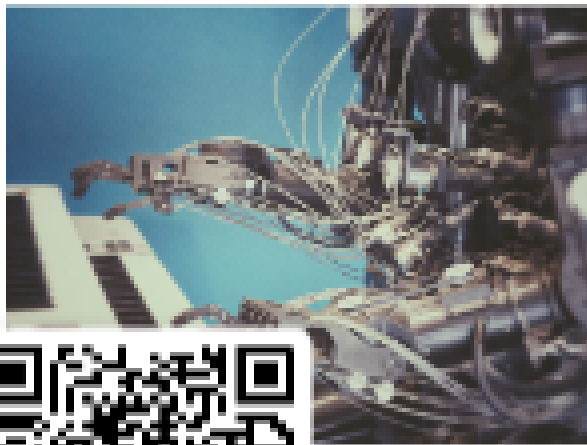

Topic 8

Intelligence

Join at menti.com | use code 2164 9632

Does Intelligence consist of one general ability or multiple specific abilities?



0

One general ability

0

Multiple specific abilities



Overview

- Introduction
- Intelligence testing
 - The psychometric approach
- Intelligence beyond the IQ test
- The relation between IQ and cognitive performance
- The roots of intelligence

Introduction

- General psychology
- Differential psychology

Introduction

- Francis Galton (1822–1911)
 - statistical analysis of mental tests.
 - Many subjects
 - Measure individual achievement on mental exercises
 - Measurement of individual differences
 - Systematical study of mental activities (motor behavior to mental imagery etc.)
 - Measured olfactory discrimination and space perception etc.

Introduction

- Pearson (1857–1936)
 - Covariation of multiple traits.
 - Mathematical possibility of predicting capacity on the basis of many tests of various mental traits.
- Spearman (1863–1945)
 - Intelligence: a single general factor and some specific traits.
- L. L. Thurstone (1887–1955)
 - Factor analysis
 - Made possible the development of a combined prediction method for individual abilities.

Introduction

- Alfred Binet (1857–1911)
 - assessing the intellectual aptitude of schoolchildren
 - the concept of mental age
- William Stern (1871–1938)
 - dividing the mental age by the chronological age to calculate an individual's intelligence quotient (IQ).
- In **1916**, re-standardized the Binet test for American use.
- In **1917**, intelligence testing for the army adopted it as a selection device

Intelligence testing

What is intelligence?

“the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly, and learn from experience”

(Gottfredson, 1997)

“the ability to use knowledge to reason, make decisions, make sense of events, solve problems, understand complex ideas, learn quickly, and adapt to environmental challenges”

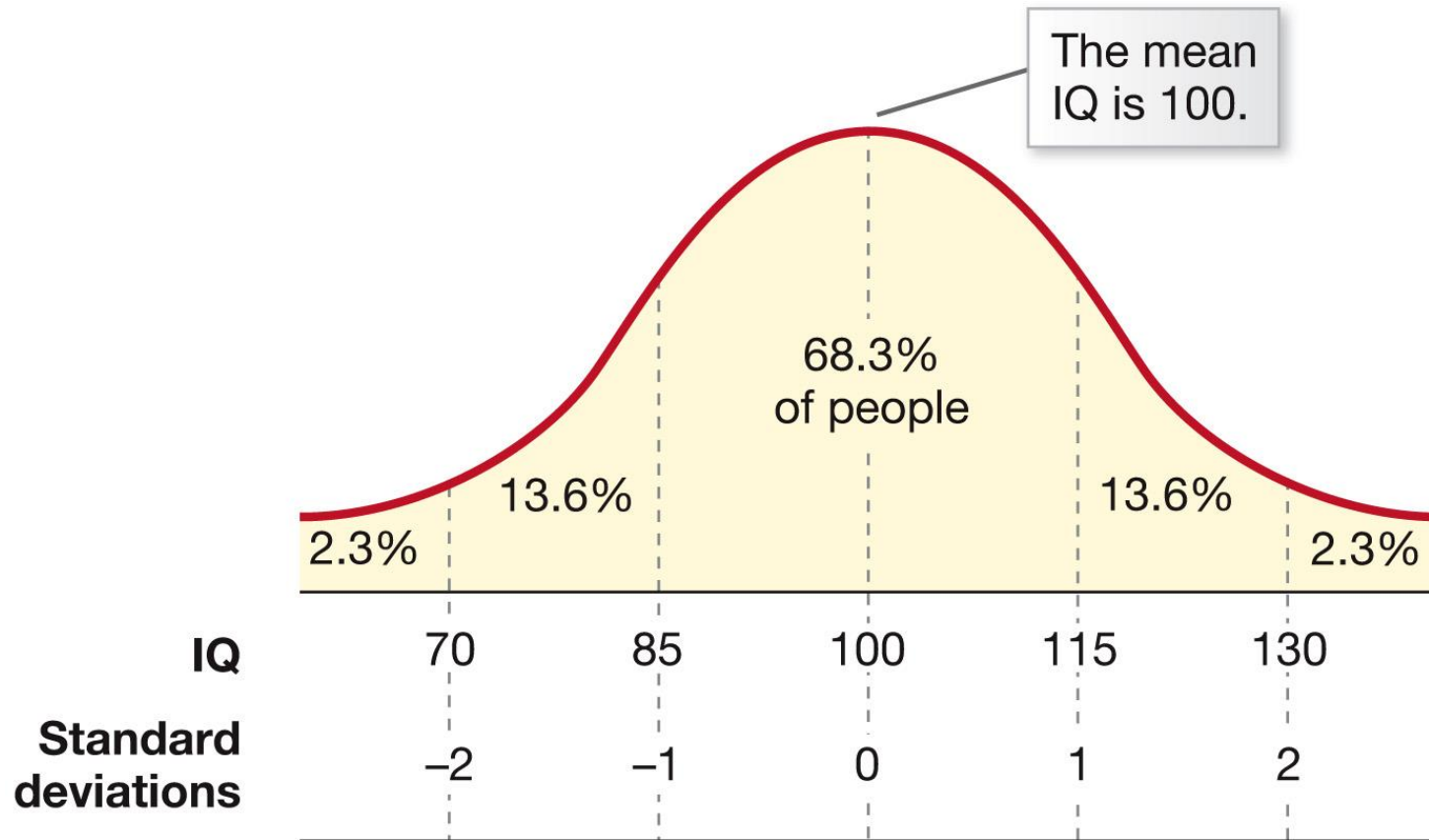
(Gazzaniga, 2018)

Intelligence testing

- How to measure intelligence?
 - Standardized psychometric tests
 - $IQ = (MA / CA) * 100$ (Binet, 1911)
MA = Mental Age
CA = Chronological Age
 - Deviation IQ



Intelligence testing



Intelligence testing

- How to measure intelligence?
 - Tests for children
 - e.g., Binet Test or Stanford-Binet Test, Wechsler Intelligence Scale for Children (WISC)



Intelligence testing

- How to measure intelligence?
 - Tests for adults
 - e.g., Wechsler Adult Intelligence Scale (WAIS / WAIS-R)
 - verbal and performance tests, individual test



Order these pictures so that they correctly tell a story.

10

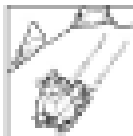


FIGURE 8.24



0
1 3 4 2

0
3 4 1 2

0
2 1 3 4

0
4 1 3 2

This is a performance task similar to those in the WAIS.

Intelligence testing

COMPREHENSION

1. Why should we obey traffic laws and speed limits?
2. Why are antitrust laws necessary?
3. Why should we lock the doors and take the keys to our car when leaving the car parked?
4. What does this saying mean: "Kill two birds with one stone."

INFORMATION

1. Who wrote *Huckleberry Finn*?
2. Where is Finland?
3. At what temperature does paper burn?
4. What is entomology?

ARITHMETIC

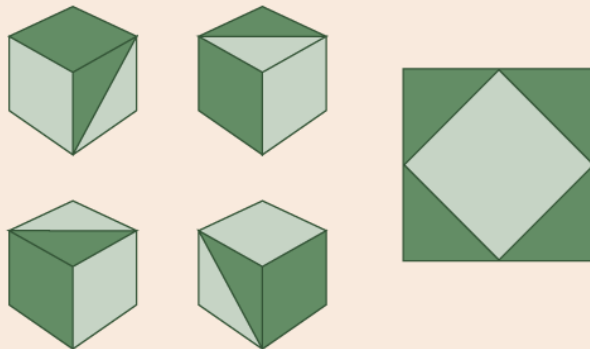
1. How many 15¢ stamps can you buy for a dollar?
2. How many hours will it take a cyclist to travel 60 miles if he is going 12 miles an hour?

A. Verbal tests

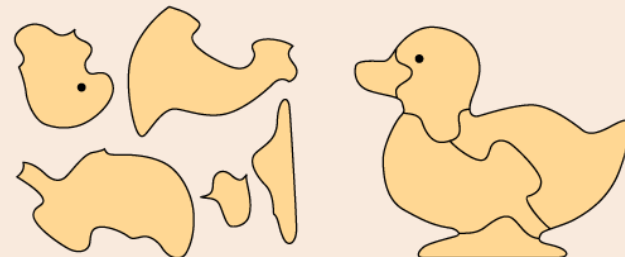
3. A man bought a used stereo system for $\frac{3}{4}$ of what it cost new. He paid \$225 dollars for it. How much did it cost new?
4. Six men can finish a job in ten days. How many men will be needed to finish the job in two and a half days?



B. Picture completion



C. Block design



D. Object assembly

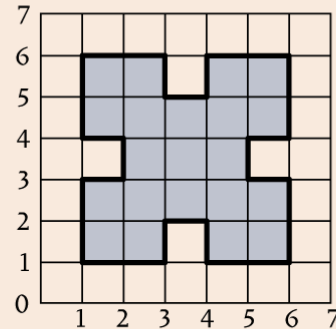
Figure 14.6: Test items similar to some in the Wechsler Adult Intelligence Scale

Intelligence testing

- How to measure intelligence?
 - Tests for adults
 - e.g., Wechsler Adult Intelligence Scale (WAIS / WAIS-R)
 - verbal and performance tests, individual test
 - e.g., Scholastic Aptitude Test (SAT)
 - group test

Intelligence testing

A



5. In the figure above, what is the area of the shaded region?

- (A) 21 (B) 24 (C) 25 (D) 28 (E) 32

B

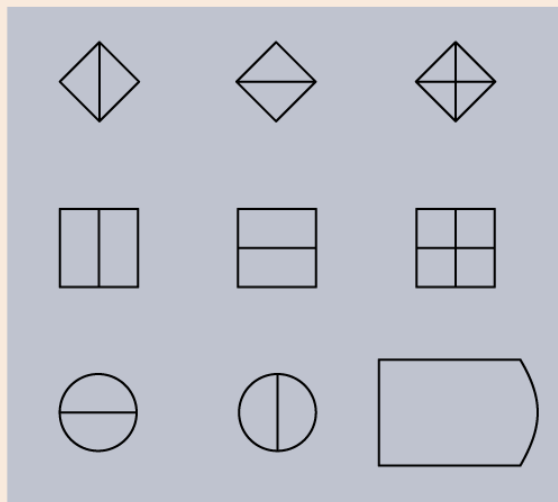
25. LINGUISTICS:LANGUAGE::

- (A) statistics:sociology
(B) ceramics:clay
(C) gymnastics:health
(D) dynamics:motion
(E) economics:warfare

Intelligence testing

- How to measure intelligence?
 - Tests for adults
 - e.g., Wechsler Adult Intelligence Scale (WAIS / WAIS-R)
 - verbal and performance tests, individual test
 - e.g., Scholastic Aptitude Test (SAT)
 - group test
 - e.g., Raven's Progressive Matrices Test
 - Non-verbal test

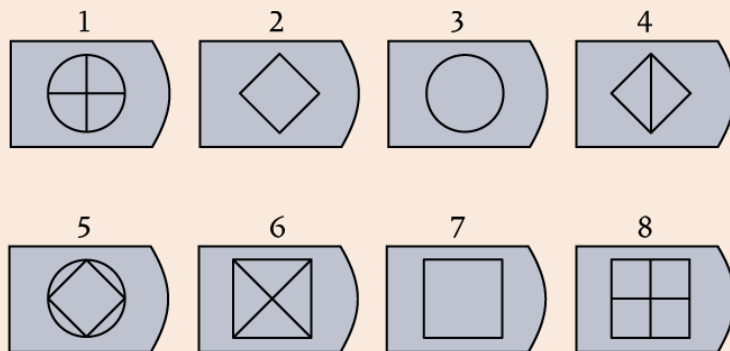
Intelligence testing



1

2

3



Intelligence testing

- Intelligence tests
 - high reliability
 - high predictive validity (but depends on the complexity of someone's job!)
- with caveats!

Intelligence testing

What is intelligence?

“the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly, and learn from experience”

(Gottfredson, 1997)

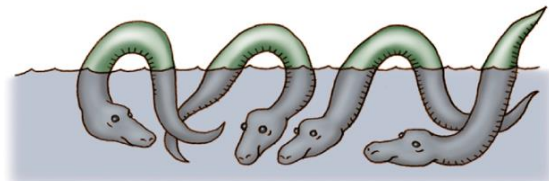
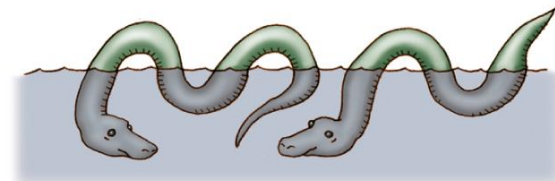
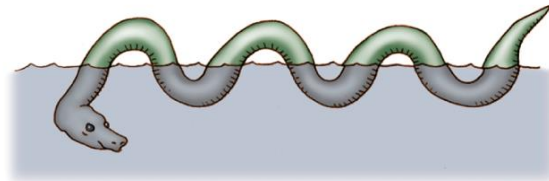
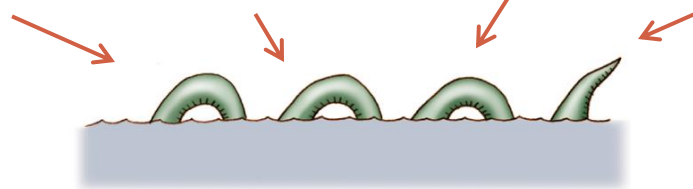
- one general ability?
- different subtypes of abilities?

The psychometric approach

- An attempt to understand the nature of intelligence by studying the pattern of results obtained on intelligence tests.
 - One general ability or multiple specific abilities?
 - e.g., *WAIS-R*

The psychometric approach

Comprehension *Information* *Arithmetic* *Vocabulary*



The psychometric approach

TABLE 14.3 CORRELATION MATRIX OF FOUR SUBTESTS ON THE WECHSLER ADULT INTELLIGENCE SCALE

	I	C	A	V
I (Information)	—	.70	.66	.81
C (Comprehension)		—	.49	.73
A (Arithmetic)			—	.59
V (Vocabulary)				—

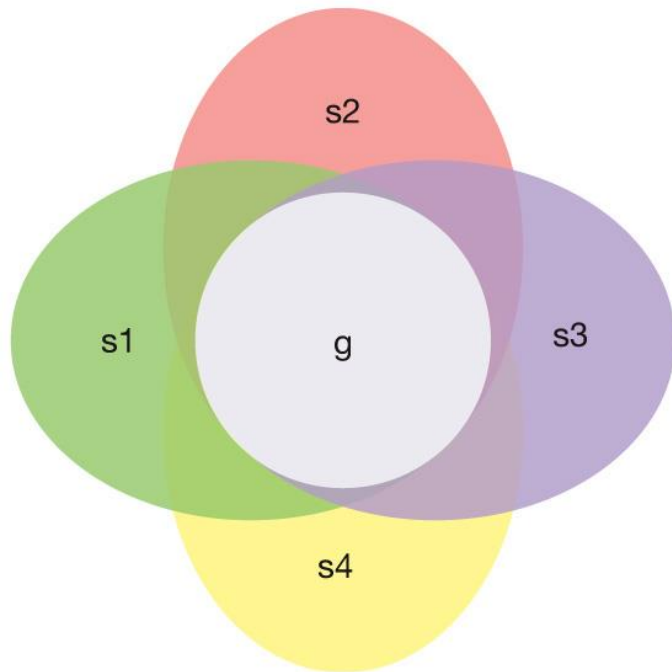
NOTE: The matrix shows the correlation of each subtest with each of the other three. Note that the left-to-right diagonal (which is here indicated by the dashes) cannot have any entries because it is made up of the cells that describe the correlation of each subtest with itself. The cells below the dashes are left blank because they would be redundant.

SOURCE: From Wechsler, 1958.

The psychometric approach

- An attempt to understand the nature of intelligence by studying the pattern of results obtained on intelligence tests.
- One general ability or multiple specific abilities?
 - e.g., *WAIS-R*
 - Spearman (1904): **general intelligence (g) + specific abilities**

The psychometric approach



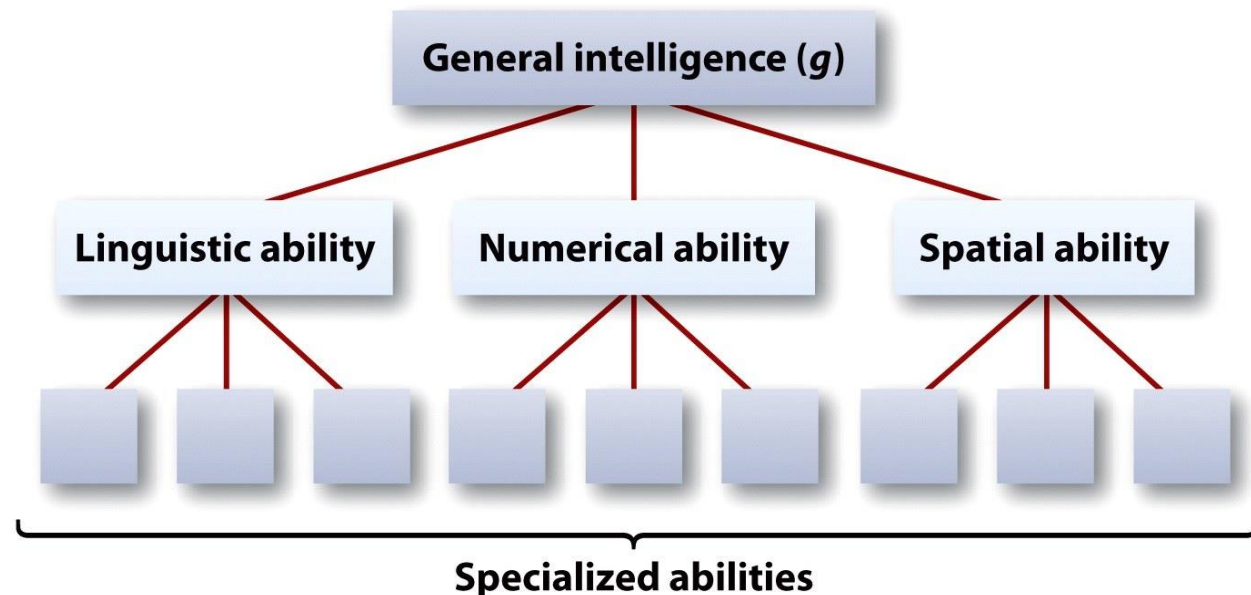
g = general intelligence

s1 = a specific ability
(e.g., math)

s2 = a second specific ability
(e.g., writing)

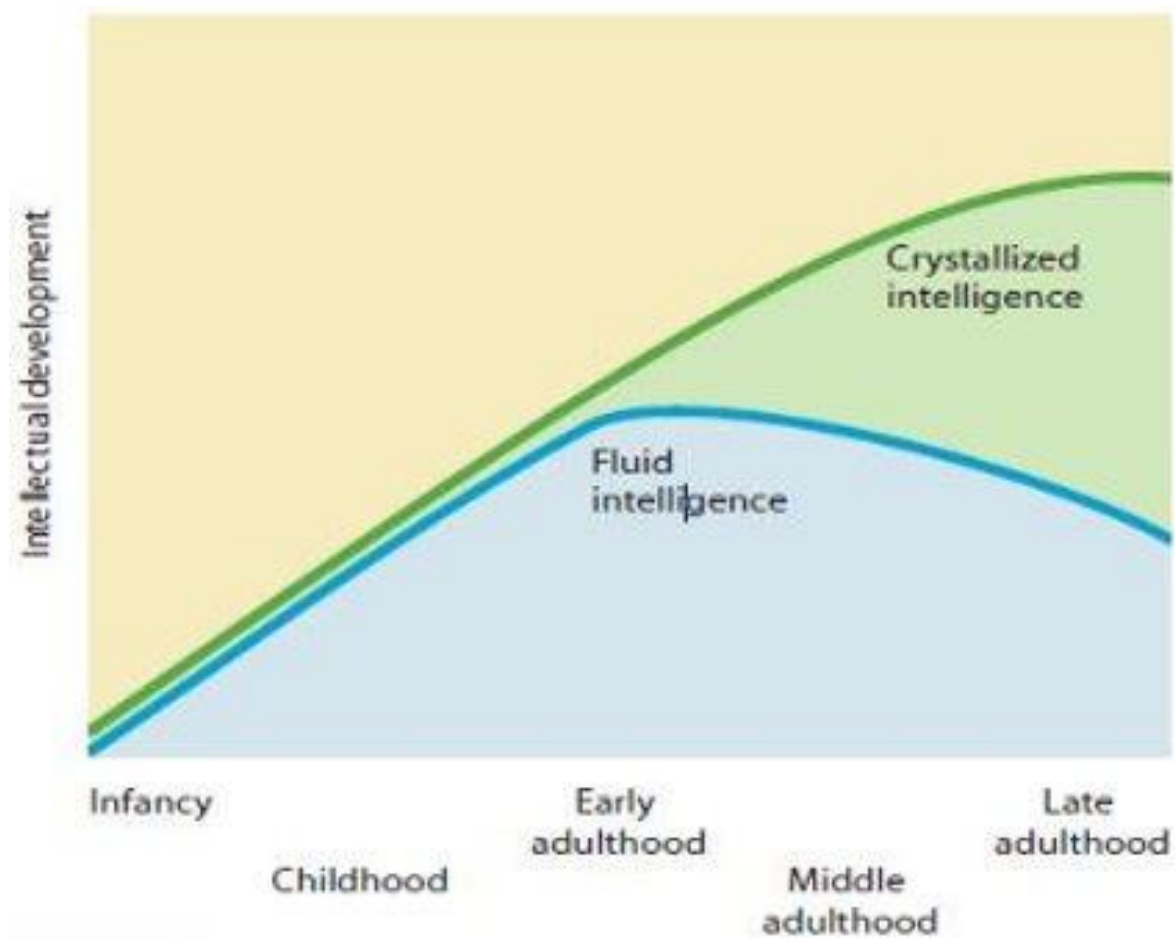
s3 = a third specific ability
(e.g., problem solving)

s4 = a fourth specific ability
(e.g., drawing)



The psychometric approach

- An attempt to understand the nature of intelligence by studying the pattern of results obtained on intelligence tests.
 - One general ability or multiple specific abilities?
 - e.g., *WAIS-R*
 - Spearman (1904): general intelligence (g) + specific abilities
 - Cattell (1971): fluid vs. crystallized intelligence



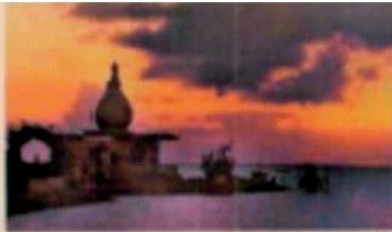
Intelligence beyond the IQ test

- Standard intelligence tests typically fail to capture “other types of intelligence”
 - Sternberg (1999): analytic, creative, and practical intelligence
 - Salovey & Mayer (1990): emotional intelligence

Intelligence beyond the IQ test

Tom felt anxious and became a bit stressed when he thought about all the work he needed to do. When his supervisor brought him an additional project, he felt _____.

- a. overwhelmed
- b. depressed
- c. ashamed
- d. self-conscious
- e. jittery



INSTRUCTIONS: How much is each feeling expressed by this picture?

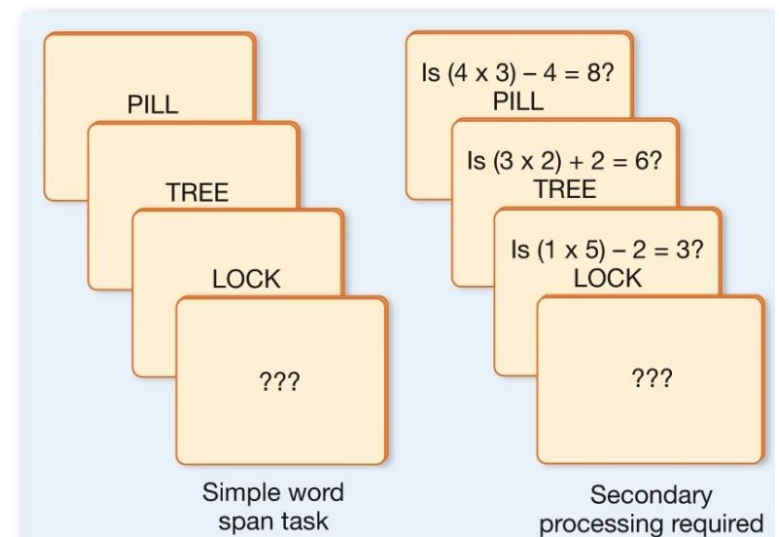
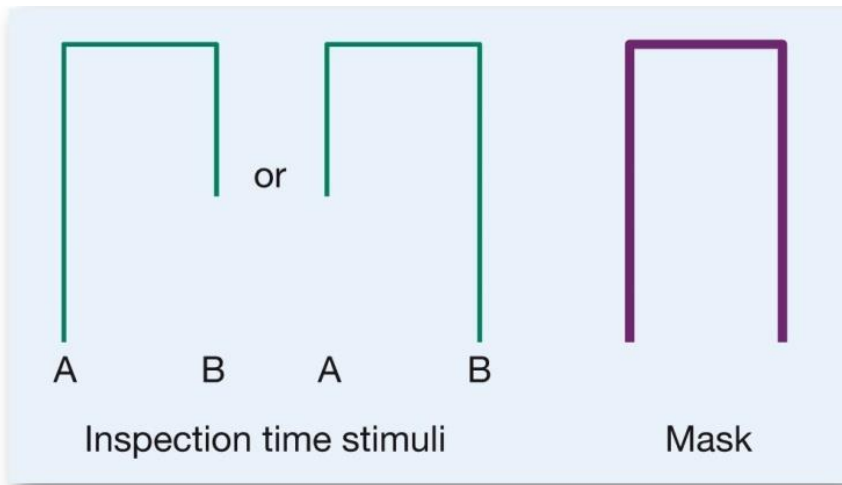
- | | 1 | 2 | 3 | 4 | 5 |
|--------------|---|---|---|---|---|
| 1. Happiness |  |  |  |  |  |
| 2. Sadness |  |  |  |  |  |

Intelligence beyond the IQ test

- Standard intelligence tests typically fail to capture “other types of intelligence”
 - Sternberg (1999): analytic, creative, and practical intelligence
 - Salovey & Mayer (1990): emotional intelligence
 - Gardner (1983): multiple intelligences
 - linguistic intelligence
 - logical-mathematical intelligence
 - spatial intelligence
 - musical intelligence
 - bodily-kinesthetic intelligence
 - inter- and intrapersonal intelligence
 - naturalistic intelligence

The relation between IQ and cognitive performance

- Different measures correlate with g
 - Simple RT and choice RT task
 - Discrimination task performance
 - Working Memory Capacity
 - Executive control



Can you cheat intelligence tests?

The authors conclude that working memory training programs appear to produce short-term, specific training effects that do not generalize to measures of "real-world" cognitive skills.

These results seriously question the practical and theoretical importance of current computerized working memory programs as methods of training working memory skills.

Melby-Lervåg, M., Redick, T. S., & Hulme, C. (2016). Working memory training does not improve performance on measures of intelligence or other measures of "far transfer" evidence from a meta-analytic review. *Perspectives on Psychological Science*, 11(4), 512-534.

The roots of intelligence

The score on an intelligence test is affected by both genes and the environment

(genes x environment)

The roots of intelligence

- Genetic influences
 - Twin studies (identical twins versus fraternal twins)



Correlation IQ: .86



Correlation IQ: .60

The roots of intelligence

- Genetic influences
 - Twin studies (identical twins versus fraternal twins)
 - Adoption studies

TABLE 14.4 CORRELATION BETWEEN THE IQS OF FAMILY MEMBERS

Identical twins reared together	+.86
Fraternal twins reared together	+.60
Siblings reared together	+.47
Child and biological parent by whom child is reared	+.42
Child and biological mother separated from the child by adoption	+.31
Child and unrelated adoptive mother	+.17

SOURCE: Data on twins, siblings, and children reared with biological parents from Bouchard and McGue, 1981; data on adopted children from Horn, Loehlin, and Willerman, 1979.

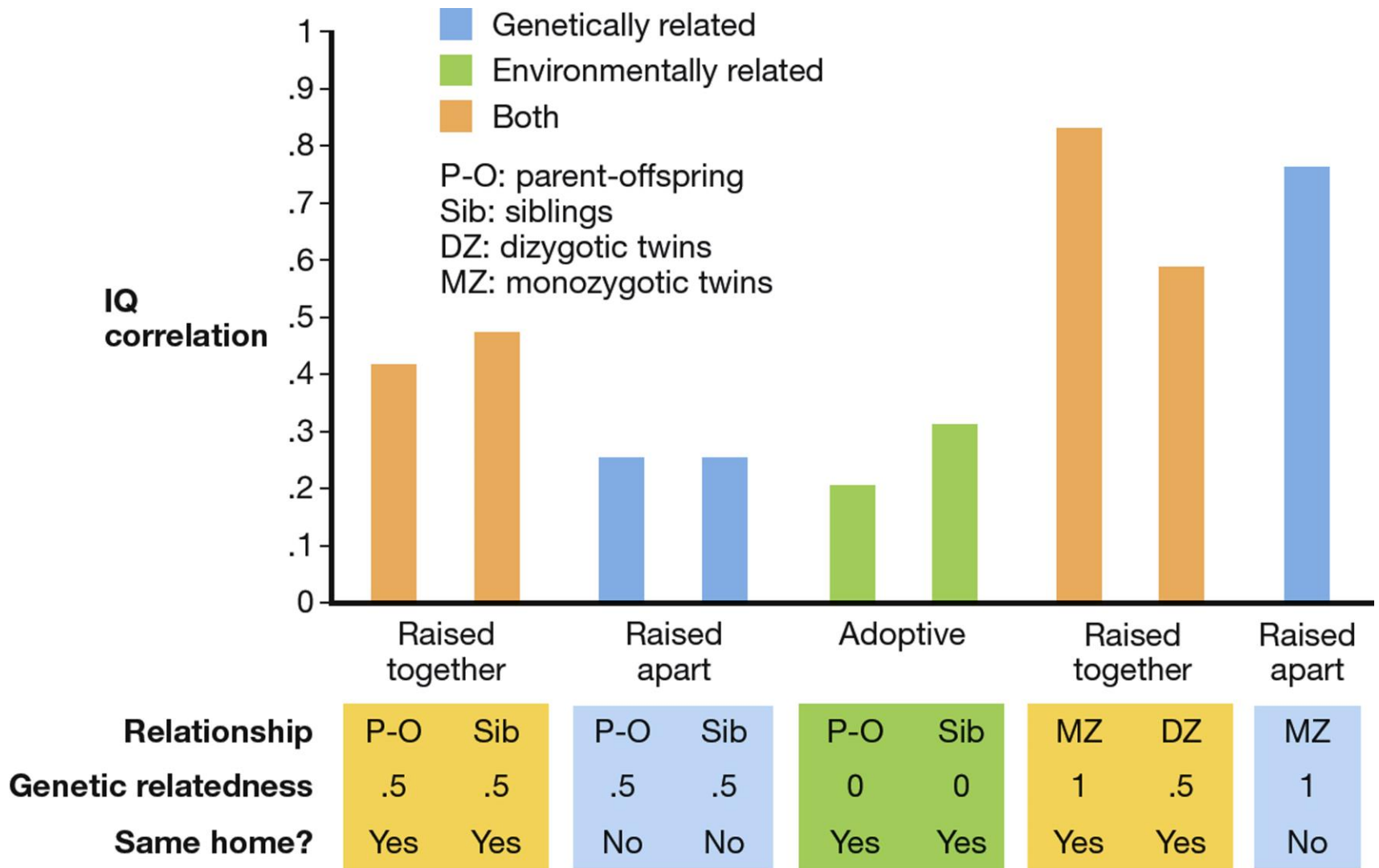
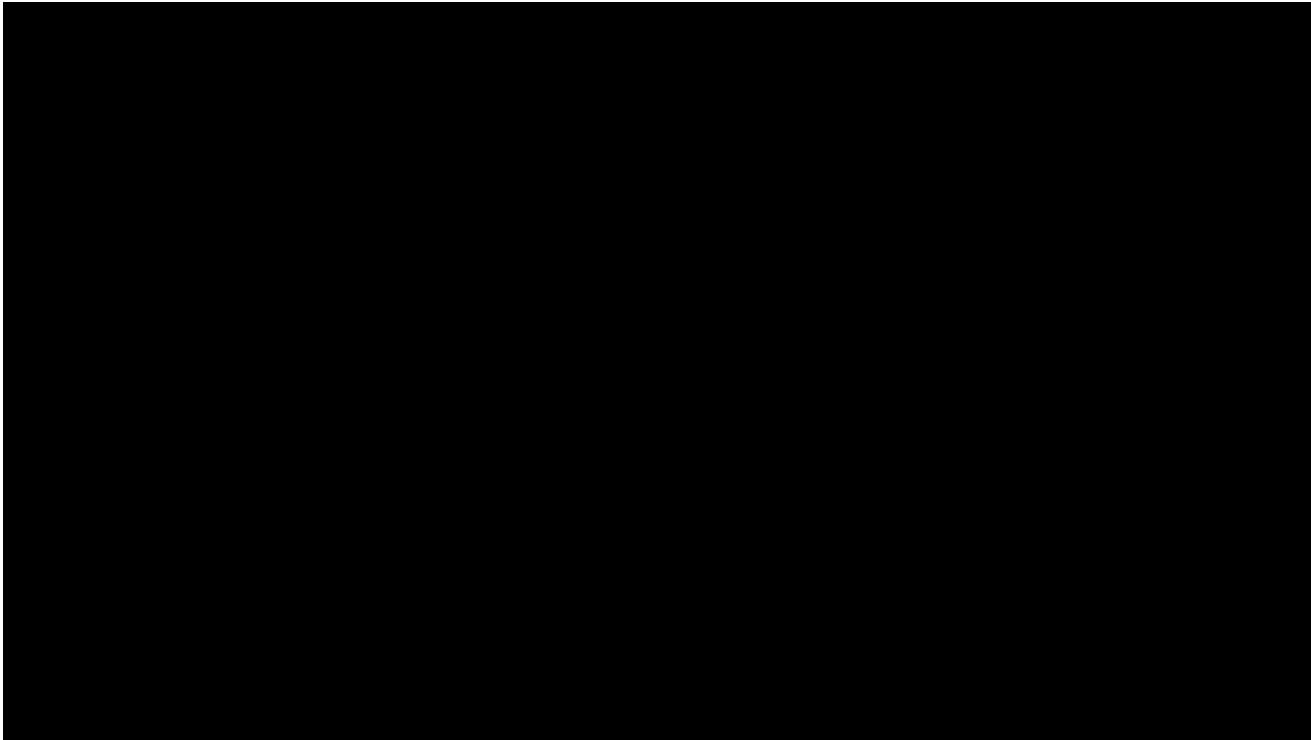


FIGURE 8.31

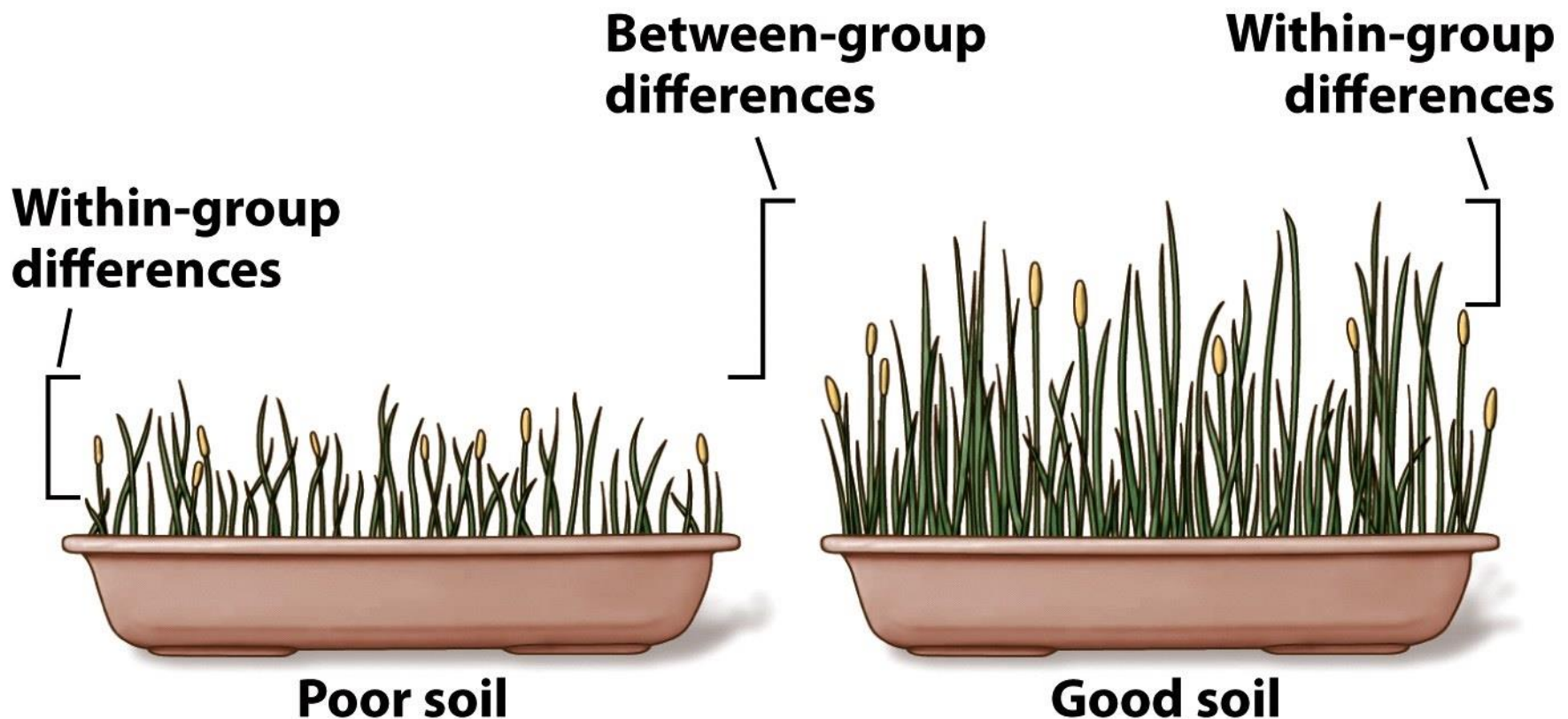
Adopted / Separated Twins

- Investigators have also studied identical twins who were separated early in life and reared apart
 - same genetics, different environment

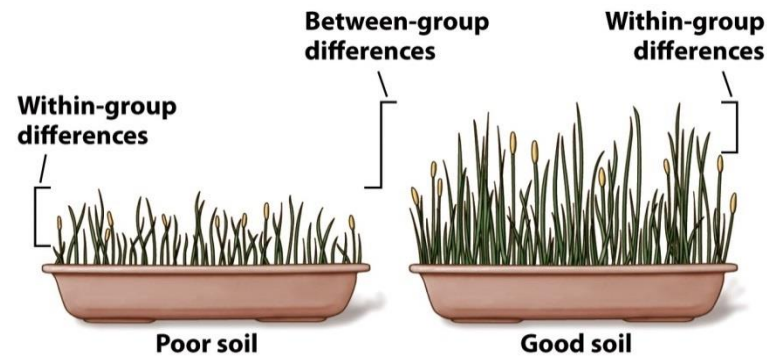


The roots of intelligence

- Environmental influences
 - Impoverished versus enriched environments

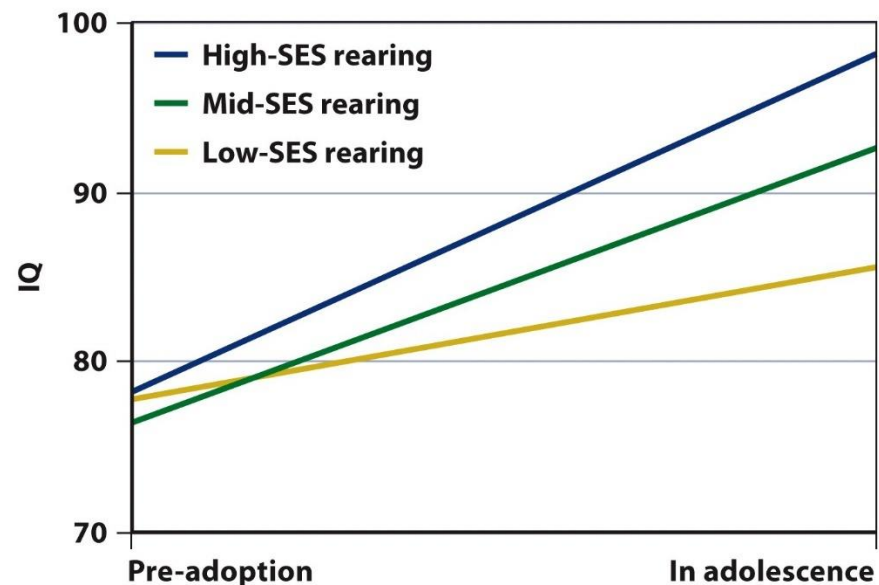


The roots of intelligence



The roots of intelligence

- Environmental influences
 - Impoverished versus enriched environments
 - Environmental factors (prenatal & postnatal factors, e.g., SES, education, overall change in environment)



The roots of intelligence

- *Heritability (H)*
 - A number between 0 and 1

$$H = \frac{\textit{Genetic Variance}}{\textit{Phenotypic Variance}}$$

$H = 0 \rightarrow$ everything is due to environment

$H = 1 \rightarrow$ everything is due to genetics (having 2 eyes)

Nature or Nurture: Estimate of Heritability

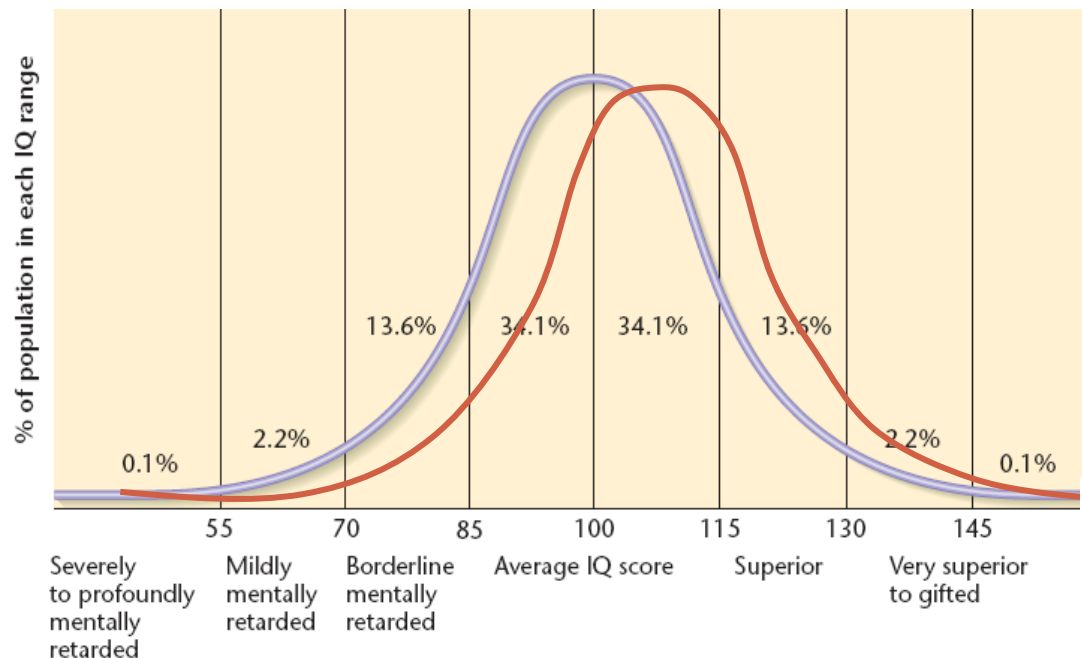
- Heritability

- A ***statistical estimate*** of the proportion of **variance** in a given population that is due to genetics.
- Heritability of .6 (between 0 and 1) = genetic influence explains about 60 percent of the observed variation among people
- Some variables such as eye color and height are highly heritable, other variables such as eating habits is less heritable.

- Heritability is not fate. It's not inevitable
 - If you change the environment, you can change the trait's heritability.
 - BMI is heritable, but access to quality food and exercise have an impact.
- High heritability does not mean group differences are genetic.
 - The capacity for having a large vocabulary is considerably heritable, but every word in a person's vocabulary is learned in an environment.

The roots of intelligence

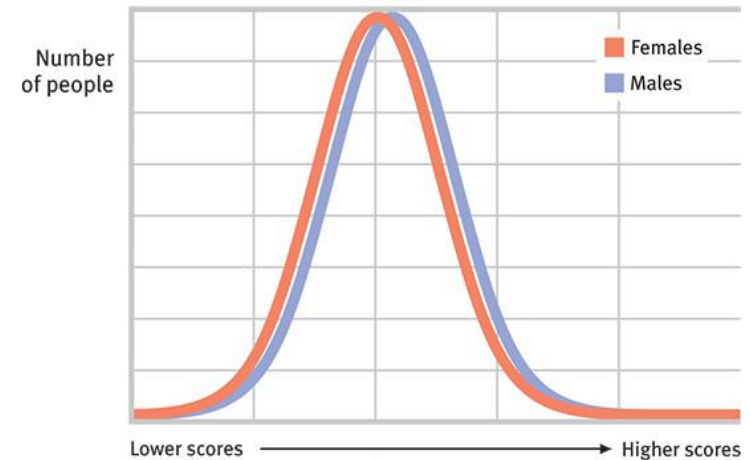
- Environmental influences
 - Impoverished versus enriched environments
 - Environmental factors (prenatal & postnatal factors, e.g., SES, education, overall change in environment)
 - Flynn effect



Reverse Flynn effect: Looking for Flynn effects in a recent online U.S. adult sample: Examining shifts within the SAPA Project

The roots of intelligence

- Group differences
 - Men versus women



Research into human IQ — long one of the most controversial areas of science — has produced a new set of ~~suitably awkward~~ results.

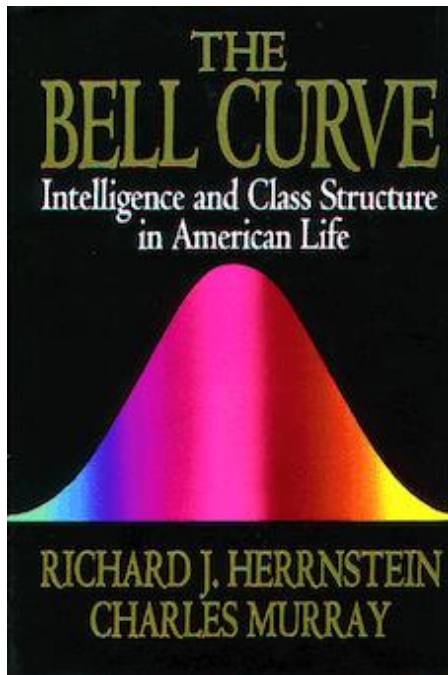
Men's average IQ is four points above women's because they typically have bigger brains.

Scientists used the latest ~~scanning techniques~~ to measure the brain volumes of 896 people, who were also subjected to a battery of intelligence tests.

“We found that the average IQ of men was about four points above that of women,” said Professor Dimitri van der Linden, of Erasmus University in Rotterdam.

The roots of intelligence

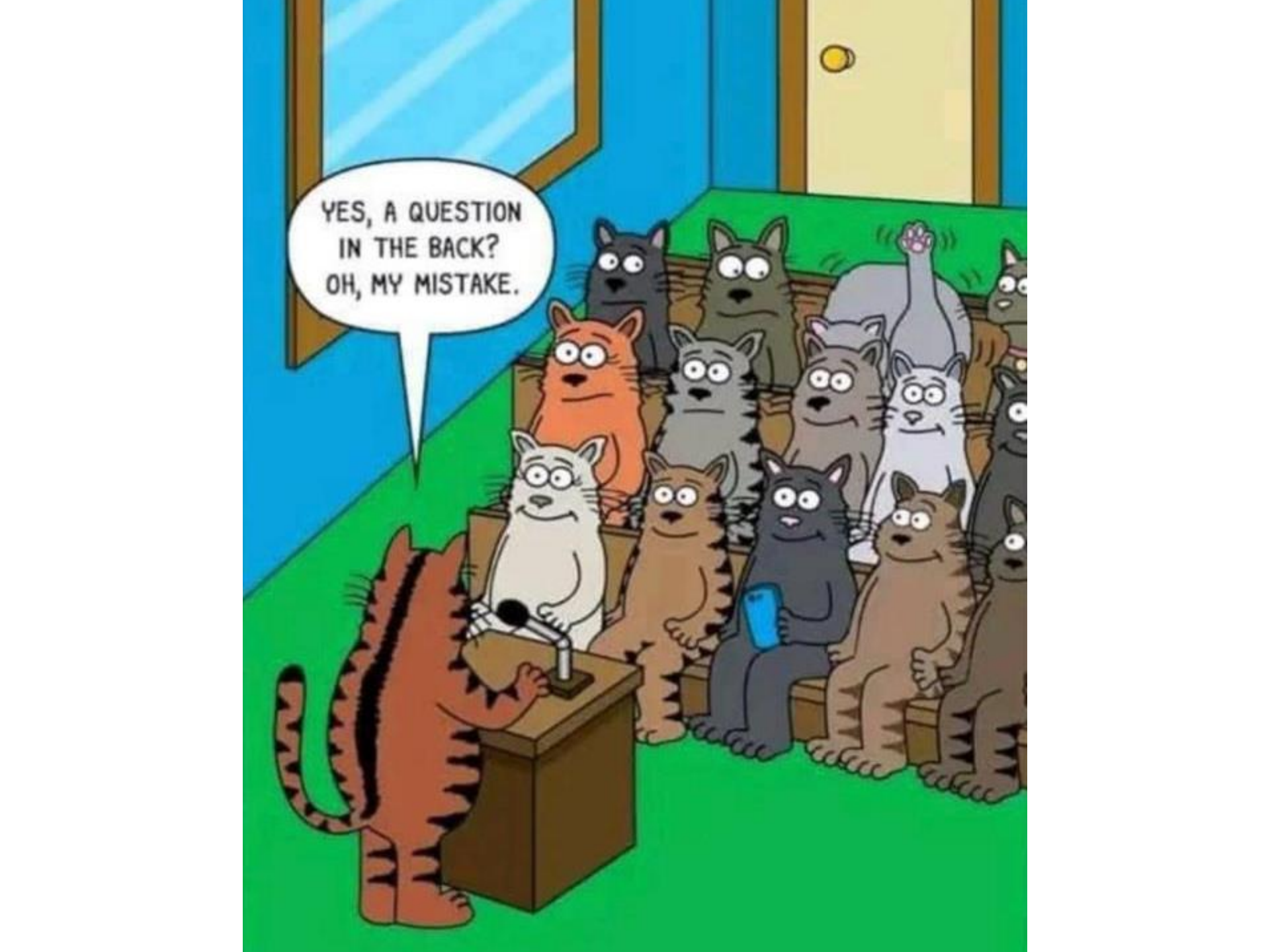
- Group differences
 - Men vs women
 - Black vs white



BUT:

APA (American Psychological Association)
special task force:

“There is certainly no such support for a genetic interpretation. At present, no one knows what causes this differential.”

A cartoon illustration of a tiger giving a presentation to a group of cats. The tiger, with orange fur and black stripes, stands at a wooden podium on the left, holding a microphone. A speech bubble from the tiger contains the text: "YES, A QUESTION IN THE BACK? OH, MY MISTAKE." The audience, consisting of various cats of different colors (white, orange, grey, brown, black), is seated in rows of wooden chairs. Most cats have wide-eyed, attentive expressions. One cat in the back row is raising its paw. The setting is a room with blue walls, a green floor, and a yellow door in the background.

YES, A QUESTION
IN THE BACK?
OH, MY MISTAKE.